Jian Liang

Room HW 106M, Haking Wong Building, The University of Hong Kong
jliang97@hku.hk • +852 9418 4970 • Homepage
Google Scholar • GitHub • LinkedIn • ResearchGate

EMPLOYMENT

The University of Hong Kong — Hong Kong, China
Postdoctoral Fellow, Department of Data and Systems Engineering — Sep 2026–Present
Supervised by Prof. Max Z.J. Shen and Prof. Shaochong Lin

EDUCATION

The University of Hong Kong — Hong Kong, China
Ph.D. in Transportation Engineering — Sep 2022–Aug 2026
Supervised by Prof. Jintao Ke

Central South University — Changsha, China
M.S. in Transportation Engineering — Sep 2019–Jun 2022
Supervised by Prof. Jinjun Tang

Central South University — Changsha, China
B.E. in Transportation Engineering — Sep 2015–Jun 2019

EXPERIENCE

University of Toronto — Toronto, Canada
Visiting Student in Rotman School of Management — Aug 2025–Jan 2026
Supervised by Prof. Sheng Liu

Victoria University of Wellington — Wellington, New Zealand
Visiting Student in School of Engineering and Computer Science — Dec 2023–Feb 2024
Supervised by Prof. Aaron Chen

The University of Hong Kong — Hong Kong, China
Research Assistant in Department of Civil Engineering — Jan 2022–Jun 2022
Supervised by Prof. Jintao Ke

The Hong Kong Polytechnic University — Hong Kong, China
Visiting Student in Department of Logistics and Maritime Studies — Aug 2021–Dec 2021
Supervised by Prof. Jintao Ke and Prof. Hongbo Ye

Huawei 2012 Lab — Shenzhen, China
Research Intern in Service Lab of Central Research Institute — Jun 2020–Sep 2020
Supervised by Dr. Hao Xu

RESEARCH INTERESTS

Methodologies

1. Data-Driven Optimization
2. Econometric Modeling & Causal Inference
3. Machine Learning & Deep Learning

Application Areas

1. On-Demand Delivery Operations
2. Travel Behavior Modeling
3. Travel Demand Forecasting

AWARDS & HONORS

Postgraduate Scholarship, The University of Hong Kong — 2022–2026
Outstanding Graduate, Hunan Province — 2022
National Scholarship for Postgraduates, China — 2021
Mittal Outstanding Student, Central South University — 2019

PUBLICATIONS

* Corresponding author. Underlined names indicate supervisors during my master's, doctoral, and visiting studies.

Working Papers

[1] **Liang, J.**, Ke, J., Lim, S., and Liu, S. (2026). "Bundling with Dynamic Claiming Behaviors: Beyond Static Design Principles." *Under review.*

[2] **Liang, J.**, Zhao, Y., Xu, J., and Ke, J. (2026). "Pricing Across Space and Time: Traffic Reallocation and Congestion Effects of Hong Kong's Cross-Harbour Toll Adjustment." *Under review.*

[3] Xu, J., Zhao, Y., **Liang, J.**, and Ke, J. (2026). "Remote Work and the Disappearing Near-Rail Rent Premium: Evidence from U.S. Rental Markets." *Under review.*

Published in Journals

[18] Wang, J., Chen, T., Shen, Z., **Liang, J.**, Zhou, B., and Ke, J. (2026). "SmartSim: An open-source mesoscopic urban traffic simulator for multimodal and reinforcement learning applications." *ENGINEERING Management.* Published online.

[17] **Liang, J.**, Feng, S., Ke, J., and Yang, H. (2025). "Machine learning for on-demand ride services: A survey." *Artificial Intelligence for Transportation*, 2, 100008.

[16] Chen, T., **Liang, J.**, Zhao, Y., & Ke, J. (2025). "To grab or not to grab? Revealing determinants of drivers' willingness to grab orders under the broadcasting mode." *Travel Behavior and Society*, 41, 101093.

[15] **Liang, J.**, Zhao, Y., Wang, H., Yang, L., and Ke, J. (2025). "Understanding order cancellation behavior in on-demand delivery services." *Transportation Research Part A: Policy and Practice*, 198, 104515.

[14] **Liang, J.** and Ke, J. (2025). "A Markov decision process framework for order dispatching in on-demand delivery services." *Multimodal Transportation*, 100240.

[13] Li, Z., Gao, F., Hao, J., **Liang, J.**, Han, C., and Tang, J. (2025). "Exploring spatiotemporal characteristics of ride-hailing ridership connecting with metro stations: A comparative analysis of holidays, weekdays, and weekends." *Journal of Transport Geography*, 123, 104104.

[12] **Liang, J.**, Zhao, Y., Wang, H., Xiao, Z., and Ke, J. (2024). "Uncovering users' willingness to wait in on-demand food delivery markets." *Transport Policy*, 158, 14-28.

[11] Gao, F., He, S. Y., Han, C., and **Liang, J.** (2024). "The impact of shared mobility on metro ridership: The non-linear effects of bike-sharing and ride-hailing services." *Travel Behaviour and Society*, 37, 100842.

[10] Gao, F., Han, C., Yang, L., **Liang, J.***, He, X., Li, F. (2024). "Analyzing spatiotemporal distribution patterns of metro ridership: comparison between common-class and business-class carriage service." *Journal of Transport Geography*, 115, 103798.

[9] **Liang, J.**, Ke, J., Wang, H., Ye, H., and Tang, J. (2023). "A Poisson-Based Distribution Learning Framework for Short-Term Prediction of Food Delivery Demand Ranges." *IEEE Transactions on Intelligent Transportation Systems*, 24(12), 14556-14569.

[8] **Liang, J.**, Tang, J., Gao, F., Wang, Z., and Huang, H. (2023). "On region-level travel demand forecasting using multi-task adaptive graph attention network." *Information Sciences*, 622, 161-177.

[7] Zhang, Y., Zou, Y., Tang, J., and **Liang, J.** (2022). "Long-term prediction for high-resolution lane-changing data using temporal convolution network. "*Transportmetrica B: transport dynamics*, 10(1), 849-863.

[6] **Liang, J.**, Tang, J., Liu, F., and Wang, Y. (2021). "Combining individual travel preferences into destination prediction: A multi-module deep learning network." *IEEE Transactions on Intelligent Transportation Systems*, 23(8), 13782-13793.

[5] Tang, J., **Liang, J.**, Liu, F., Hao, J., and Wang, Y. (2021). "Multi-community passenger demand prediction at region level based on spatio-temporal graph convolutional network." *Transportation Research Part C: Emerging Technologies*, 124, 102951.

[4] Tang, J., **Liang, J.**, Yu, T., Xiong, Y., and Zeng, G. (2021). "Trip destination prediction based on a deep integration network by fusing multiple features from taxi trajectories." *IET Intelligent Transport Systems*, 15(9), 1131-1141.

[3] Yang, X., Zou, Y., Tang, J., **Liang, J.**, and Ijaz, M. (2020). "Evaluation of short-term freeway speed prediction based on periodic analysis using statistical models and machine learning models." *Journal of Advanced Transportation*, 2020(1), 9628957.

[2] Tang, J., **Liang, J.**, Han, C., Li, Z., and Huang, H. (2019). "Crash injury severity analysis using a two-layer Stacking framework." *Accident Analysis & Prevention*, 122, 226-238.

[1] Tang, J., **Liang, J.**, Zhang, S., Huang, H., and Liu, F. (2018). "Inferring driving trajectories based on probabilistic model from large scale taxi GPS data." *Physica A: Statistical Mechanics and its Applications*, 506, 566-577.

PATENTS

[1] Tang, J., **Liang, J.**, Han, C., and Huang, H. "Road traffic accident occurrence risk determination method and system." CN111126868B, 2023/07/04.

TEACHING EXPERIENCE

Teaching Assistant

CIVL7019 Statistical Methods for Civil Engineering, The University of Hong Kong — Fall 2022, 2023, and 2024

MATH1853 Linear Algebra, Probability and Statistics, The University of Hong Kong — Spring 2023, 2024, 2025, and 2026

CIVL3120 Transportation Infrastructure Engineering, The University of Hong Kong — Spring 2026

Research Supervision of Students

[1] Donglin Jia, Master thesis (2023), "Estimation of User Waiting Time for Online Food Delivery Service Based on Deep Recurrent Survival Analysis".

[2] Wong Jun Yan Johnson, Master thesis (2023), "Investigation of public policy measures to promote green transportation in Hong Kong".

[3] Yuanhao Xu, Master thesis (2023), "Short-term Travel Demand Prediction with Artificial Intelligence".

[4] Fuwen Deng, Master thesis (2024), "Survey Study for Food Delivery Customers' Cancellation Triggers".

[5] Wentao Meng, Master thesis (2024), "Survey Study for Influence of Communication/Technology Interventions on Food Delivery Customers' Cancellation Behaviour".

[6] Yize Zhao, Master thesis (2025), "Examining Factors Affecting Food Delivery Drivers' Routing Behavior".

[7] Bowen Ma, Master thesis (2025), "Investigating Food Delivery Drivers' Order Response Behavior".

[8] Jiyan Li, Master thesis (2025), "Predicting Food Delivery Order Arrival Times Using Deep Neural Networks".

[9] Chan Ki Yuen, Final year project (2026), "Analyzing the impact of estimated arrival time and delivery fee on customer ordering behavior".

[10] Ho Lok Weng, Master thesis (2026), "Analyzing the impact of heat hazards on driver behavior in ride-hailing services".

[11] Chen Jiaying, Master thesis (2026), "Accurate estimation of order arrival time in food delivery services".

PROFESSIONAL SERVICES

Service in Academic Organizations

Student Committee Chair for HKU Institute of Transport Studies Sept 2024–Aug 2025

Referee for Academic Journals

- Applied Energy
- Cell Reports Sustainability
- Cities
- Computers & Industrial Engineering
- Computers & Operations Research
- Decision Analytics Journal
- Engineering Applications of Artificial Intelligence
- Expert Systems With Applications
- Environmental Pollution
- IET Intelligent Transport Systems
- Neurocomputing
- Journal of Advanced Transportation
- Journal of Retailing and Consumer Services
- Journal of Traffic and Transportation Engineering
- Journal of Transport Geography
- Knowledge-Based Systems
- PLOS ONE
- Research in Transportation Economics
- Research in Transportation Business & Management
- Service Science
- Travel Behaviour and Society
- The Journal of Supercomputing
- Transport Policy
- Transportmetrica A: Transport Science
- Transportmetrica B: Transport Dynamics
- Transportation Science
- Transportation Research Part C: Emerging Technologies
- Transportation Research Part E: Logistics and Transportation Review
- Transportation Research Part A: Policy and Practice

Referee for Peer-reviewed Conferences

- International Symposium on Transport Network Resilience (INSTR)
- International Conference of Hong Kong Society for Transportation Studies (HKSTS)
- COTA International Conference of Transportation Professionals (CICTP)
- World Transport Convention (WTC)

PROJECTS

[1] Postgraduate Independent Exploration and Innovation Project, Central South University, "Passenger demand forecasting based on multi-task spatio-temporal graph neural network at region level". June 2021–June 2022, RMB 10,000, PI.

[2] RGC Early Career Scheme (ECS) 2023–2024, HKU27203323, "Modelling, simulation, and operational strategy designs for on-demand food delivery services", 1 Jan 2024–31 Dec 2026, Core Member.

[3] Smart Traffic Fund, PSRI/78/2311/RA, "SmartSim: AI-assisted Simulation Software for Multimodal Transportation Operations", 1 Sep 2024–31 Aug 2026, Core Member.

CONFERENCE PRESENTATION

[1] **Liang J.** and Ke J., 2024. Order assignment for on-demand food delivery service considering driver behavior. Proceedings of *the 28th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2024)*, Hong Kong, China, December 9–10, 2024.

[2] **Liang J.** and Ke J., 2024. Order assignment for on-demand food delivery service considering driver behavior, Presented in *the 15th International Workshop on Computational Transportation Science (CTS)*, Chengdu, China, June 29–30, 2024.

[3] **Liang J.** and Ke J., 2023. A Poisson-based distribution learning framework for short-term prediction of food delivery demand ranges. Proceedings of *the 9th International Symposium on Transport Network Resilience (INSTR2023)*, Hong Kong, China, Dec 13–14, 2023.

[4] **Liang J.** and Ke J., 2023. Understanding users' cancellation behaviors in online food delivery markets: A reference-dependent survival analysis. Proceedings of *the 27th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2023)*, Hong Kong, China, December 11–12, 2023.

[5] **Liang J.** and Ke J., 2023. Understanding users' cancellation behaviors in online food delivery markets: a reference-dependent survival analysis, Presented in *the World Transport Convention (WTC 2023)*, Wuhan, China, June 14–17, 2023.

[6] **Liang J.**, Ke J., Wang H., Ye H., and Tang J, 2023. A Poisson-Based Distribution Learning Framework for Short-Term Prediction of Food Delivery Demand Ranges. Presented in *the 102nd Annual Meeting of Transportation Research Board (TRB 2023)*, Washington DC, United States, January 8–12, 2023.

[7] **Liang J.**, Ke J., Ye H., Wang H. and Tang J., 2022. A Poisson-Based Distribution Learning Framework for Short-Term Prediction of Food Delivery Demand Ranges. Proceedings of *the 26th International Conference of Hong Kong Society for Transportation Studies (HKSTS 2022)*, Hong Kong, China, December 12–13, 2022.

[8] Wang, J., Chen, T., Shen, Z., **Liang, J.**, Zhou, B., and Ke, J., 2025. SmartSim: A Scalable and Multimodal Open-Source Mesoscopic Urban Traffic Simulator. Presented in *the 29th International Conference of Hong Kong Society for Transportation Studies, HKSTS 2025: Smart Mobility*, Hong Kong, China, December 8–9, 2025.

[9] **Liang, J.**, Wang, C., Chen, T., and Ke, J. 2026. Adaptive Order Dispatching with Cost Estimation and Postponement Learning, Presented in *the 105th Transportation Research Board (TRB) Annual Meeting*, Washington DC, United States, January 11–15, 2026.

[10] Chen, T., **Liang, J.**, Zhao, Y., and Ke, J., 2026. To grab or not? Revealing determinants of drivers' willingness to grab orders in on-demand ride services, Presented in *the 105th Transportation Research Board (TRB) Annual Meeting*, Washington DC, United States, January 11–15, 2026.

REFERENCES

Prof. Max Z.J. Shen
Chair Professor in Logistics and Supply Chain Management
Department of Data and Systems Engineering
The University of Hong Kong
Email: maxshen@hku.hk

Prof. Jintao Ke
Assistant Professor
Department of Civil Engineering
The University of Hong Kong
Email: kejintao@hku.hk

Prof. Hai Wang
Associate Professor
School of Computing and Information Systems
Singapore Management University
Email: haiwang@smu.edu.sg

Prof. Sheng Liu
Assistant Professor
Rotman School of Management
University of Toronto
Email: sheng.liu@rotman.utoronto.ca